Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

Bab 1

PENDAHULUAN

Bab 2

TINJAUAN PUSTAKA

[12pt,english]report [T1]fontenc [latin9]inputenc babel longtable float calc amsmath amsthm setspace [unicode=true,pdfusetitle, bookmarks=true,bookmarksnumbered=false,bookmarksopen=false, breaklinks=false,pdfborder=0 0 1,backref=false,colorlinks=false] hyperref

UTSAthesis times latexsym

definition plain

subfig

First Name Last Name, Ph.D.

First Name Last Name, Ph.D.

First Name Last Name, Ph.D.

First Name Last Name, Ph.D.

First Name Last Name, Ph.D.

Doctor of Philosophy in Computer SciencePh.D.M.Sc.Department of Computer ScienceCollege of SciencesDecember 2016

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I would like to dedicate this thesis/dissertation template to UTSA graduate students.

A TEMPLATE THESIS/DISSERTATION USING THE UTSATHESIS PACKAGE FOR LaTeX AND LyX USERS

Weining Zhang (to be replaced by your own name) First of all, I would like to thank Kevin Xu Su for creating an earlier version of the LaTeX style, Lijie Zhang for using an earlier version of this package to write her dissertation and to provide feedback.

I would also like to thank the UTSA Graduate School for reviewing the outcome of this template document and correction of formatting errors. This sample is based on UTSA 2016 Dissertation/Thesis format.

(Notice: If any part of the thesis/dissertation has been published before, the following two paragraphs should be included without alteration).

This Masters Thesis/Recital Document or Doctoral Dissertation was produced in accordance with guidelines which permit the inclusion as part of the Masters Thesis/Recital Document or Doctoral Dissertation the text of an original paper, or papers, submitted for publication. The Masters Thesis/Recital Document or Doctoral Dissertation must still conform to all other requirements explained in the Guide for the Preparation of a Masters Thesis/Recital Document or Doctoral Dissertation

at The University of Texas at San Antonio. It must include a comprehensive abstract, a full introduction and literature review, and a final overall conclusion. Additional material (procedural and design data as well as descriptions of equipment) must be provided in sufficient detail to allow a clear and precise judgment to be made of the importance and originality of the research reported.

It is acceptable for this Masters Thesis/Recital Document or Doctoral Dissertation to include as chapters authentic copies of papers already published, provided these meet type size, margin, and legibility requirements. In such cases, connecting texts, which provide logical bridges between different manuscripts, are mandatory. Where the student is not the sole author of a manuscript, the student is required to make an explicit statement in the introductory material to that manuscript describing the students contribution to the work and acknowledging the contribution of the other author(s). The signatures of the Supervising Committee which precede all other material in the Masters Thesis/Recital Document or Doctoral Dissertation attest to the accuracy of this statement.

Ringkasan

The first chapter of this document is a description of the content and the usage of the UTSA-thesis package. The remaining chapters serve to illustrate some use of LyX features for writing a thesis/dissertation.

The first line of the abstract has been indented as per required by the thesis/dissertation guideline.

Bab 3

Introduction

Several different versions of LaTeX packages have been used in the past by sciences and engineering PhD students to write Doctoral Dissertations since 2008. However, these packages are not compatible with each other. Also, there has been no LyX layout for writing UTSA thesis/dissertation. The UTSAthesis package was initially created to allow my students to use LyX to write their doctoral theses.

The UTSAthesis package contains UTSAthesis.sty, a LaTeX package for LaTeX users, and utsathesis.layout, a LyX layout for LyX users. These packages provide formatting commands that confirms to the guideline of thesis/dissertation of UTSA Graduate School as of September 2011. The UTSAthesis package also contains a sample and a template of a UTSA thesis/dissertation, for both LaTeX and LyX users.

The UTSAthesis.sty package was created based on a LaTeX package written by Kevin Xu Su in 2008. A number of PhD students from the Department of Computer Science have tested the earlier version of the packages and provided comments and suggestions. The Graduate School has provided comments to ensure that the style confirms to the requirements.

To write a thesis/dissertation using LaTeX, a suitable TeX/LaTeX distribution should be installed on the computer system. There are many TeX/LaTeX distributions for various operating systems, including TeXLive for Linux/Unix, MacTex for Mac OS, and MikTeX for Windows. Each distribution contains instructions for installation and configuration. For people who want to write thesis/dissertation using LyX, an appropriate LyX package can be downloaded from www.lyx.org and installed on the system, after the installation of the TeX/LaTeX distribution. The UTSAthesis package should then be installed into appropriate LaTeX and LyX directories within the computer system.

In the rest of this Chapter, I will discuss the content and usage of the UTSAthesis.sty and utsathesis.layout packages. In subsequent chapters, I will illustrate some typical structures of a doctoral dissertation. The content in those chapters is not directly related to the use of the thesis package.

3.1 The UTSAthesis.sty Package

The UTSAthesis.sty is a LaTeX macro package that defines the LaTeX style of the doctoral dissertation and MS thesis for the University of Texas at San Antonio. It provides LaTeX commands to

format thesis title, author, abstract, copyright, dedication, acknowledgments, supervisor, committee members, degrees, department, and college.

This package should be used with the LaTeX standard report class. See the sampleThesis.tex for an example and template. Other LaTeX packages will also typically be included in the a thesis/dissertation.

3.1.1 Installation

Place the UTSAthesis.sty in a directory under your personal texmf tree, for example, in /home/yourhome/.texmf/tex/latex/thesis. Remember to run texconfig to re-hash file list, so that the LaTeX can find this style file.

3.1.2 Use of UTSAthesis.sty Package

The general structure of a typical UTSA thesis/dissertation file, such as sampleThesis.tex, should look like the follows.

```
\documentclass[12pt,english]{report}
\usepackage{UTSAthesis}
... use other packages ...
\begin{document}
\committee{... }
\informationitems{... }
\thesiscopyright{...}
\dedication{\emph{I would like to dedicate this thesis/dissertation to ...}}
\title{\textbf{First line}\\ \textbf{second line }...}
\author{...}
\maketitle
\begin{acknowledgements} ... \end{acknowledgements}
\begin{abstract} ... \end{abstract}
\newpage
\pagenumbering {arabic}
\setcounter {page}{1}
\pagestyle{plain}
\chapter{...} % or \include{chap3}
...
\singlespace
\bibliographystyle{...}
\bibliography{...}
\begin{vita}...\end{vita}
```

Gambar 3.1: Structure of a thesis LaTeX file

The following commands are defined in UTSAthesis.sty and should be used in the order suggested in Fig. 3.1 to provide required format information.

- \title{Thesis Title}. This can contain multiple lines. Use "\\" to go to the next line.

- \author{Name of Thesis Author}
- \thesiscopyright{Optional Copyright Statement}
- \dedication{Optional Dedication}
- Use the following commands to set up the dissertation/thesis committee
  - \supervisor{The first and last name, Degree}
  - \cosupervisor{The first and last name, Degree} if there is a co-supervisor. Otherwise, \committeeB{Committe member B Name, Degree}. Notice that, if co-supervisor command is used, it must be placed after the supervisor command.
  - \committeeC{Committe member C, Degree}
  - \committeeD{Committe member D, Degree}
  - \committeeE{Committe member E, Degree}
- Either \informationitems{Full Name of Degree}{Short Name of Degree}{Full Name of Department}{Full Name of College}{Month of Thesis}{Year of Thesis} or use the following commands separately.
  - \degree{Full Degree Name}
  - \degreeshort{Short Degree Name}
  - \department{Department Name}
  - \college{College Name}
  - \thesismonth{Month}
  - \thesisyear{Year}
- \maketitle is the command that produces the the title page, copyright page, and dedication page. The position of this command is critical for getting the correct formatting.
- \begin{acknowledgements}

  People, organization, supports that you want to thank for

  \end{acknowledgements}
- \begin{abstract}

  The abstract starts here. Should within one page.

  \end{abstract}
- The thesis/dissertation should then continue with chapters, appendixes, references. Before the first chapter, it is necessary to set Arabic page number. If the thesis/dissertation is long, it may be better to place chapters into separate LaTeX files and include these sub-files using \include{} command.
- \begin{vita}

  The last item is a one-page curriculum vita

  \end{vita}

3.1.3 Produce the Outcome

To produce the pdf version of the thesis/dissertation, run pdflatex and bibtex.

3.2 The utsathesis.layout Package

The utsathesis.layout is an LyX layout that provides a LyX document layout for UTSA dissertation/thesis. This layout should be used together with the UTSAthesis.sty.

3.2.1 Installation

First, install UTSAthesis.sty as described in Section 3.1. Then, installed the LyX on your system by following the instruction that comes with the LyX package. Next, place the utsathesis.layout into your personal LyX directory. On a Linux/Unix system, this directory is at ~/.lyx/layouts. On Mac OS, it is at /User/<name>/Library/Application Support/LyX-<version>/layouts. On Windows 7, it is at C:\Users\<name>\AppData\Roaming\lyx<version>\layouts. Remember to run Tools->Reconfigure inside LyX to re-configure the system.

3.2.2 Use of utsathesis.layout Package

This document (sampleThesis.lyx) provides a template for using the utsathesis.layout to write a Ph.D. dissertation. For a Master's thesis, go to Document->Settings and set the class option to ms. Other important settings may include Document->Settings->LaTeX Preamble, and the bibliography style.

The document setting should be "report (UTSAthesis 2016)". The document should begin with committee info, thesis info, copyright, and dedication. These can be formatted using items in the FrontMatter in the pull-down menu. These should be followed by title, author, acknowledgments and the abstract. The placement and the order of these four items are important for generating the correctly formatted front pages of the thesis/dissertation. It is also important to add the "Start First Page" item right before the first chapter. This item will set the correct page numbers for the main portion of the thesis/dissertation.

At the end of the document, the "Vita" item in the BackMatter in the pull-down menu needs to be used to format a one-page vita.

Regular chapters can be included in the main thesis document or more likely as sub-files, one per chapter. If sub-files are preferred, make sure the document settings of all sub-files are identical to the main document.

Bab 4

Literature Review

We have some citations [**?**, **?**, **?**]. See the Bibliography for the format of references.

Bab 5

Solution and Evaluation

In this chapter, we show the structures of math formula, theorem commands, and floats (such as algorithm and table).

5.1 A Theory

Definition 1 *This is another definition.*

Theorem 1 *This is a theorem.*

$$X = \frac{AB}{Y} \tag{5.1}$$

The proof is done here.

5.2 An Algorithm

The following is the algorithm.

5.2.1 Evaluation

The evaluation results is shown in the following table. It is straightforward to place the caption of the table above or below the table.

The following is a long table

|c|c|c|c|c| A Long Table

Column1	Column 2	Column 3	Column 4	Column 5

Algorithm 5.1 The Do-It-Yourself Method

1. Step One
2. Step Two

Tabel 5.1: Evaluation Results

	Method 1	Method 2	Method 3
Criterion 1			
Criterion 2			
Criterion 3			

Bab 6

Future Directions

There can be more chapters.

Lampiran A

Notations

Here we show the use of multiple appendixes.

A.1 Math Notations

Each appendix can have sub-sections as a regular chapter.

A.2 Additional Notations

Lampiran B

Ontologies

These is another appendix.

This should be a one-page short vita.
There can be more paragraphs.